ABHISHEK GUPTA

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience in designing and developing Generative AI and machine learning solutions for enterprise applications. Skilled in building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, RESTful APIs with FastAPI, and data-driven AI solutions using Python. Strong foundation in Exploratory Data Analysis (EDA), machine learning algorithms, CNNs, parameter-efficient model design, and transfer learning, complemented by hands-on experience in deploying scalable AI systems. Passionate about translating cutting-edge AI research into practical, high-impact solutions that drive business value.

EDUCATION

Indian Institute of Information Technology, Bhopal	M.Tech in Artificial Intelligence	9.29	2024–2026
Rajkiya Engineering College, Sonbhadra	B.Tech in Electronics Engineering	7.88	2020–2024

EXPERIENCE

Software Engineer *Kellton Tech Solutions* *May 2026 – Present*

Project: ML-Based Delivery Time Estimation *August 2026 – Present*

- Streamlined delivery data retrieval from PostgreSQL using SQLAlchemy ORM and identified predictive features through EDA.
- Optimized XGBoost hyperparameters using GridSearchCV to improve delivery-time estimation.
- Developed multivariate XGBoost models to predict three sequential delivery-time components and aggregate them into final delivery-time estimates.
- Achieved a baseline delivery compliance rate of 73%, measuring orders delivered within the promised timeframe.
- Exposed delivery-time predictions through REST APIs built with FastAPI.

Project: AI-Powered Observability Platform *June 2026 – August 2026*

- Standardized unstructured application logs by developing a preprocessing pipeline for efficient analysis.
- Enabled relevance-based log retrieval by storing embeddings in ChromaDB and implementing hybrid search with a cosine similarity threshold of 0.5.
- Improved context prioritization by reranking retrieved documents using cross-encoder/ms-marco-MiniLM-L-6-v2 before LLM-based root cause generation.
- Achieved 92% context relevance, 95% groundedness, and 92% answer relevance by evaluating the chatbot using the RAG Triad.

Project: ML-Based Utility Cost Estimation *May 2026 – June 2026*

- Identified key cost drivers through EDA to align input features with cost prediction objectives.
- Optimized cost prediction by engineering features, encoding categorical variables, and tuning XGBoost hyperparameters using Optuna.
- Developed a multivariate XGBoost model to predict four cost components simultaneously, achieving R^2 scores of 0.75 for material, 0.77 for internal labor, 0.78 for external labor, and 0.78 for other/miscellaneous costs.

Trainee *Kellton Tech Solutions* *Oct 2025 – Apr 2026*

Project: Survey Intelligence Platform *Nov 2025 – Mar 2026*

- Streamlined survey data access by integrating MongoDB with Beanie ODM for response retrieval and processing.
- Visualized survey click intensity by generating OpenCV-based heatmap images and storing them in Azure Blob.
- Summarized video audience reactions by aggregating ratings at user reaction timestamps and generating line graph.
- Exposed heatmap and video pulse analytics through REST APIs built with FastAPI.

PROJECTS

Alzheimer's Disease Prediction using Lightweight CNN [Repo Link] *July 2025 – Oct 2025*

- Developed a lightweight 3-layer CNN for four-class Alzheimer's disease classification, achieving 96.88% and 98.12% test accuracy across two MRI datasets.
- Addressed class imbalance using data augmentation and SHA-256-based duplicate detection, enhancing training data diversity and model robustness.
- Achieved 98% cross-dataset accuracy through bidirectional evaluation, demonstrating generalization.
- Validated model consistency using 5-fold cross-validation, achieving mean accuracies of 96.88% and 97.78% across the two datasets.

SKILLS

- **Programming & Backend:** Python, SQL, REST APIs
- **Machine Learning:** Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Cross-Validation, EDA, ETL Pipeline
- **Deep Learning:** Neural Networks, Convolutional Neural Networks (CNNs), Transfer Learning, Parameter-Efficient Model Design, Model Optimization
- **Natural Language Processing:** Text Preprocessing, Tokenization, Text Embeddings, Sentiment Analysis
- **Generative AI:** Retrieval-Augmented Generation (RAG), LLM Integration, Tool Calling, RAG Evaluation, AI Agents
- **Frameworks:** LangChain, LangGraph, FastAPI
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Sentence Transformers, spaCy, OpenCV, XgBoost
- **Databases & Vector Stores:** PostgreSQL, MongoDB, ChromaDB
- **Cloud & AI Platforms**: Microsoft Azure, Azure Blob Storage, Azure OpenAI, OpenAI API, Google Gemini API
- **DevOps & Development Tools**: Git, GitHub, GitHub Actions, CI/CD Pipelines, Docker, Linux, Apache Kafka

ACHIEVEMENT

- Selected among 30,000 applicants nationwide for **Amazon ML Summer School 2025**.

RESEARCH WORK

- Lightweight CNN Architecture for Efficient Alzheimer's Disease Classification
- MITRA: Trust-Aware Hardware IP Design Framework with Mutual IP Protection of Vendor and Buyers during High-Level Synthesis

CERTIFICATIONS

- **Generative AI Certification** - *Oracle* [Certificate Link] *Sep 2025*
- **AI Foundations Associate** - *Oracle* [Certificate Link] *Sep 2025*
- **Data Science using Python** - *Prutor @ IIT Kanpur* [Certificate Link] *Oct 2022*